Hearth: Prototype Evidence

Neighbourhood help on a live map, in Bengali, Hindi and English

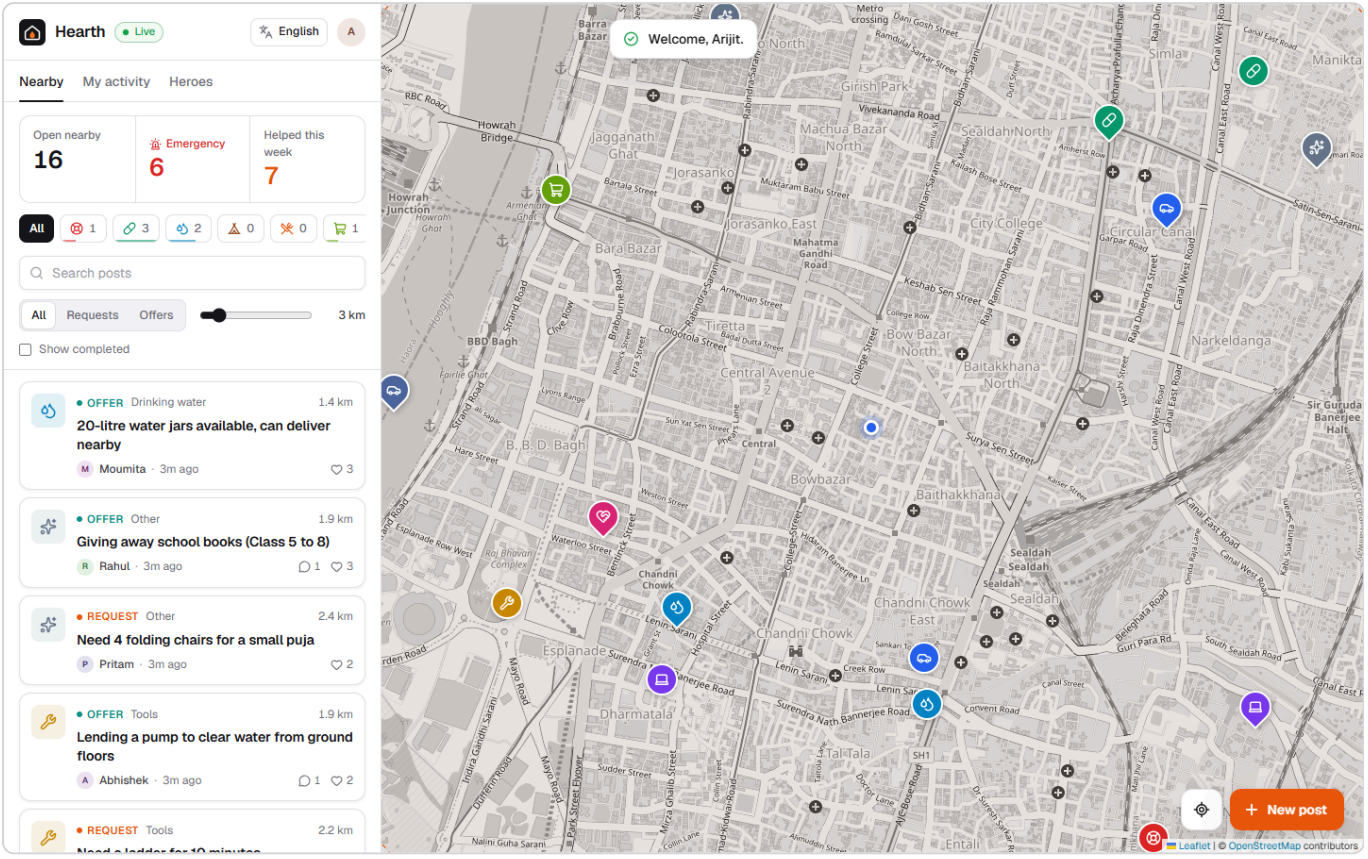
Seva First Innovation Challenge 2026
Eastern Region · Disaster Management & Community Safety

Status: working software prototype, publicly available.
Source code: github.com/sanjaysaini383/seva-first
Demo area: central Kolkata, with realistic flood-relief and everyday posts.
Applicant: Sanjay Saini (individual)

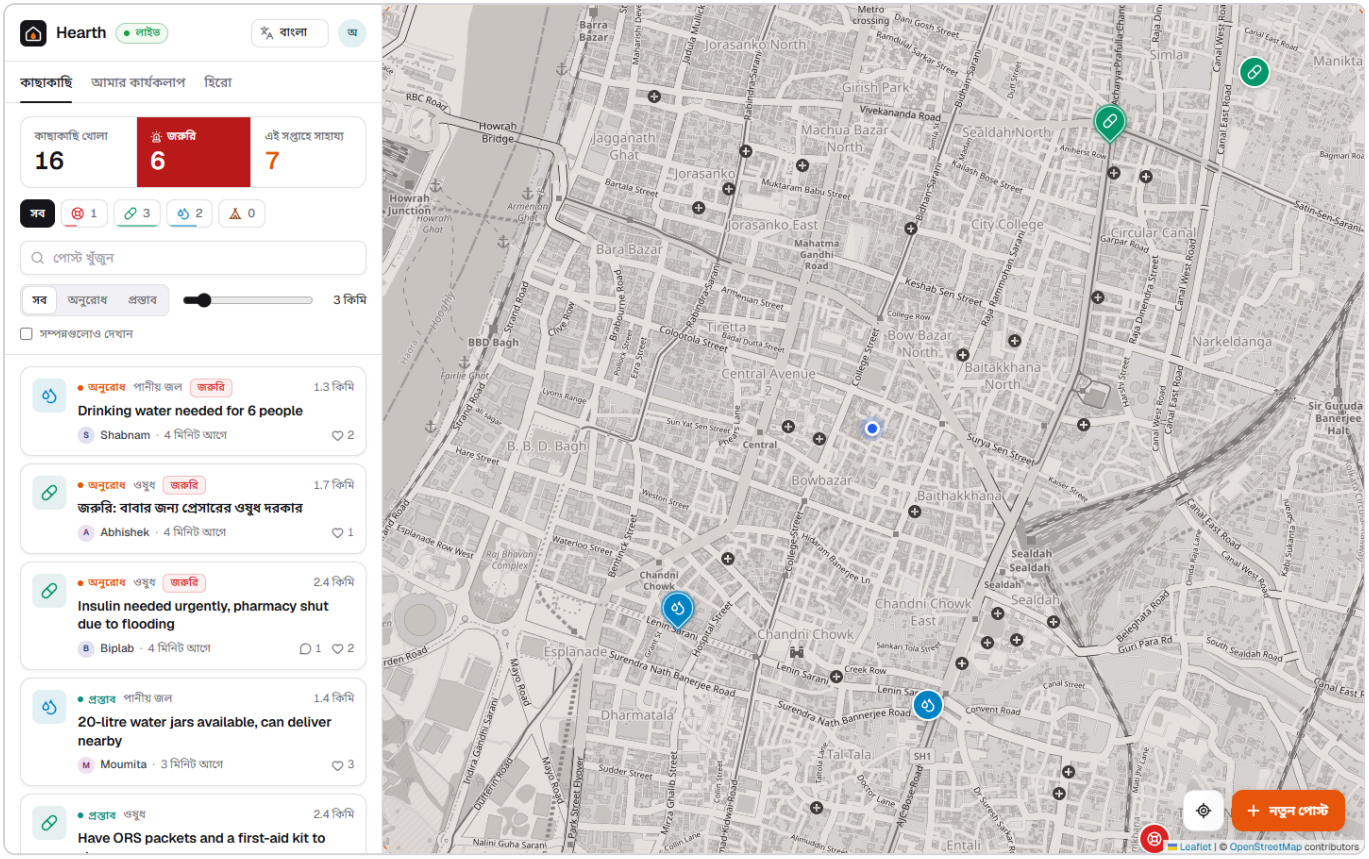
What the prototype does

- **Live map** of requests and offers for help near the user, with an adjustable search radius.
- **Emergency categories first:** Rescue / boat, Medicine, Drinking water, Shelter. **Emergency mode** shows only open emergency posts, most urgent first.
- **"I can help"** claiming: only one person can claim a request, so nobody is double-booked.
- **Conversation** on each post to coordinate details; **mark as done** with thanks.
- **Heroes board** recognising the most active volunteers.
- **Bengali, Hindi and English** interface, switchable at any time; posts can be written in any language.
- **No install or password:** opens in a phone browser; users join with a first name only.

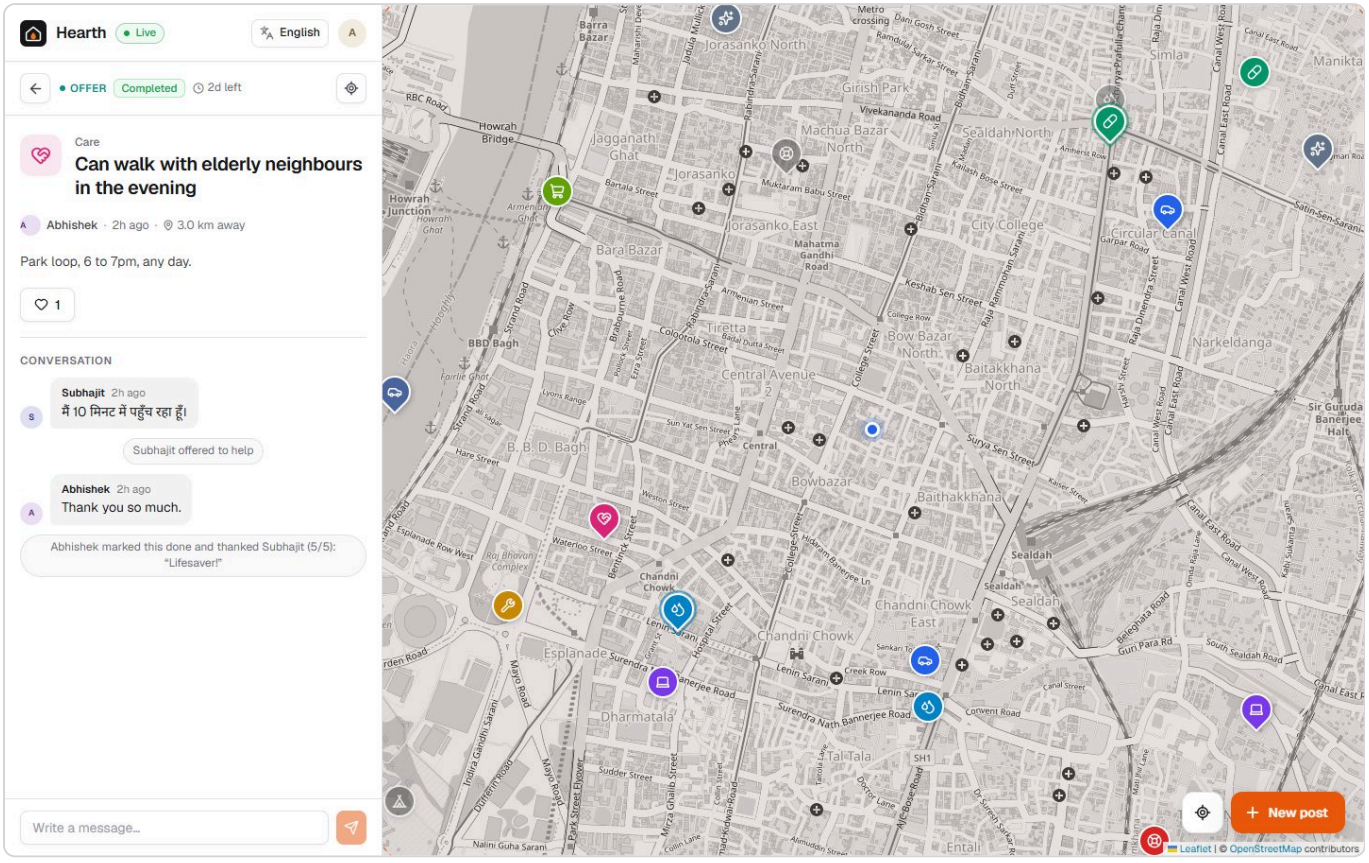
Screens from the working prototype



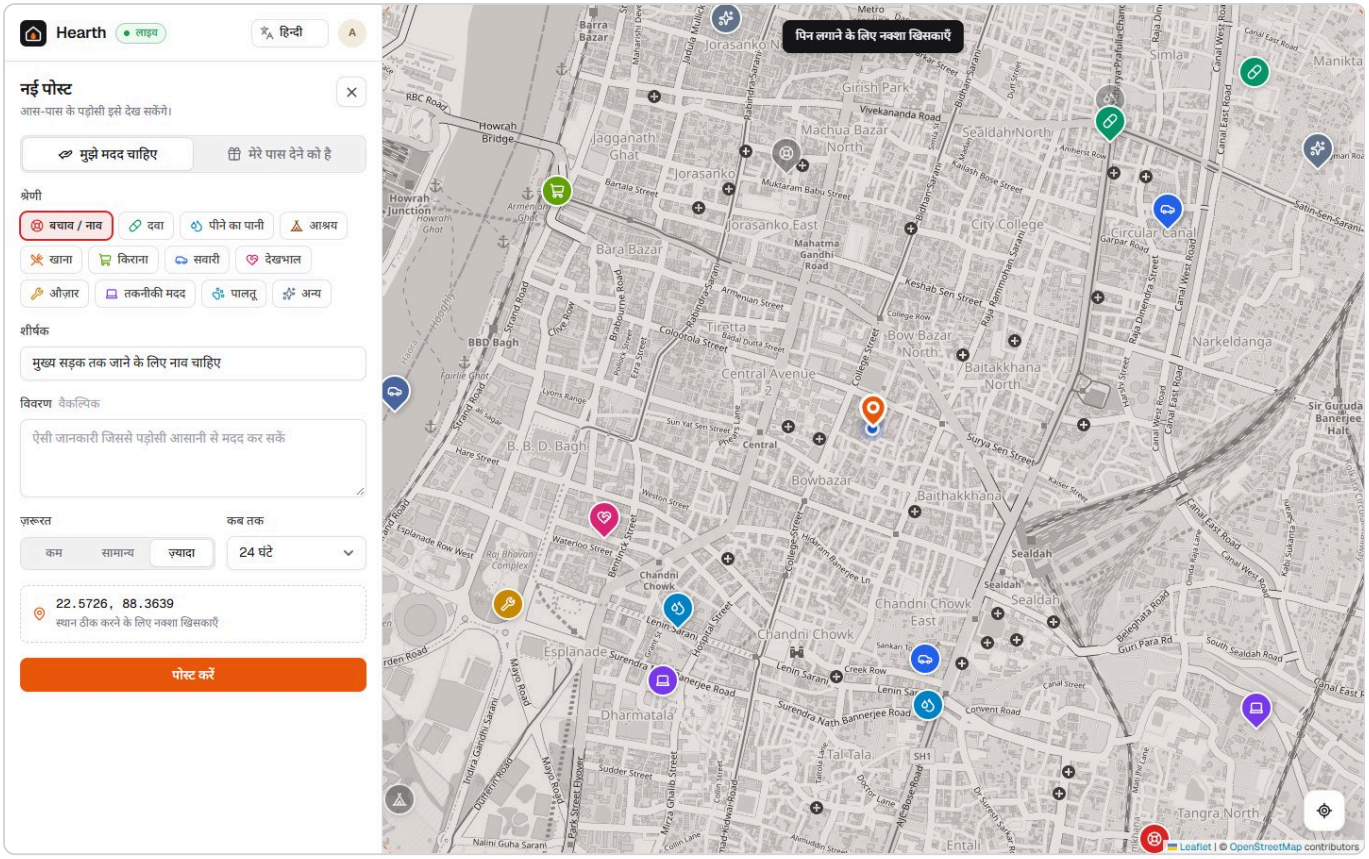
1. Main screen: open requests and offers around central Kolkata, live counts (open nearby, emergency, helped this week) and category filters.



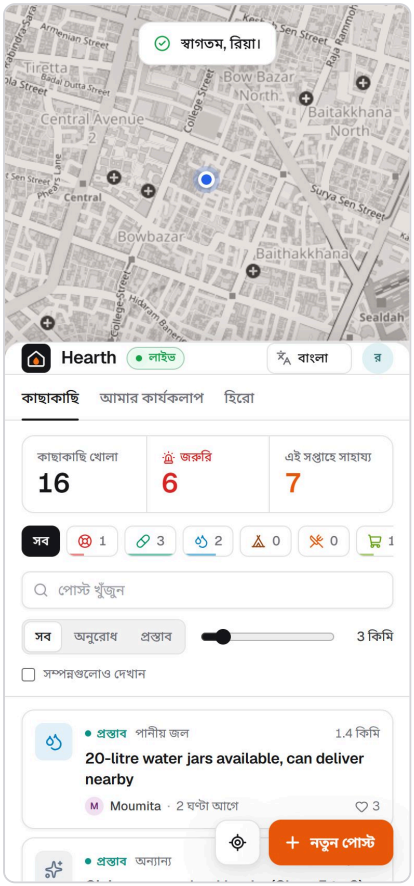
2. Emergency mode in the Bengali interface: only urgent medicine, water, rescue and shelter posts are shown.



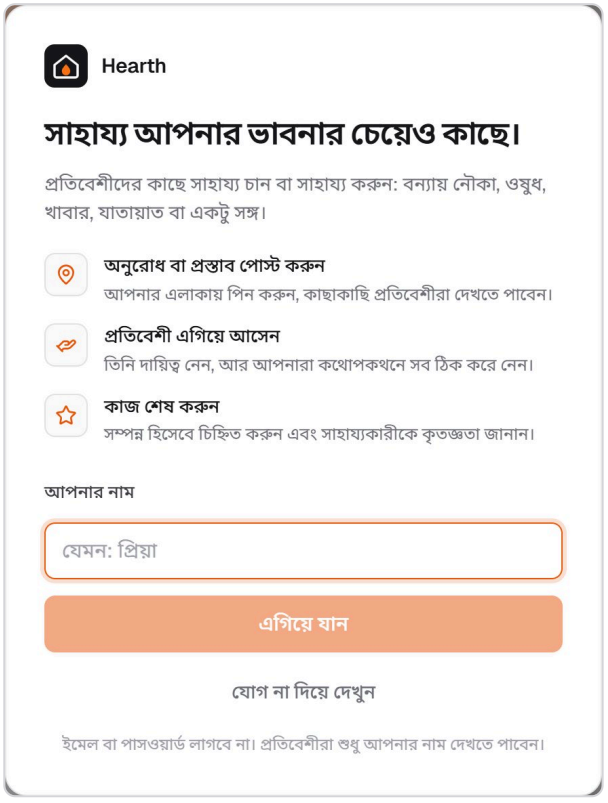
3. A completed post: a neighbour offered help, they coordinated in the conversation (including a reply in Hindi), and the post was marked done with thanks.



4. Creating a request in the Hindi interface: choose a category (Rescue / boat selected), set urgency, and drag the map to place the pin.



5. On a phone, in Bengali.



6. Joining in Bengali: first name only, no email or password.

Testing

An automated test (`npm run smoke`) checks every core function against the live cloud database. Latest run: **13 of 13 passed**.

#	What is checked	Result
1	Find posts within a chosen radius	**Pass** · Radius search returned 21 posts within 3 km and 41 within 20 km
2	Show open and in-progress posts, newest first	**Pass** · 25 posts returned in the correct order
3	Search post titles	**Pass** · Matching posts found
4	Load a post together with its conversation	**Pass** · 4 messages loaded in one request
5	Count open posts per category (live counts)	**Pass** · Counts calculated by the database
6	Total thanks received per helper (Heroes board)	**Pass** · Totals calculated by the database
7	"My activity": posts I created or am helping with	**Pass** · Correct posts returned
8	Count help completed this week	**Pass** · 7 completed posts counted
9	Appreciation counter with no lost updates	**Pass** · Increment and decrement applied safely
10	Claim a request in a single transaction	**Pass** · Post moved to "in progress" with a system message
11	Two people claim the same request at once	**Pass** · Second claim rejected (409 conflict); first helper kept
12	Mark done: status, thanks and message saved together	**Pass** · All three saved in one step
13	Delete a post	**Pass** · Its conversation is removed too; nothing left behind

A separate **two-user browser test** confirmed the full flow between two people: posting, the request appearing instantly on the other screen, claiming, chatting, marking done with thanks, the Heroes board, and deleting a post.

Current stage

The software prototype is complete. **No field pilot has been run yet**; the proposal is to pilot it in one flood-prone Kolkata ward with local volunteers through a monsoon season (see *implementation_plan.pdf*).

Technology: React and TypeScript web app; Appwrite Cloud (open source) for database, sign-in and live updates; Leaflet with OpenStreetMap maps. No paid APIs.